

CSMFAB000000000002 — Ancient Hearth: Ceramic Component Fabrication Plan

Version 2.0 — Revised & Expanded | June 2026

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Δ Change Log — V1.0 → V2.0

Rev	Date	Change Description
1.0	2024	Initial release — $\text{ZrB}_2/\text{Si}_3\text{N}_4$ composite burner, gel-casting, GPS sintering, Voronoi geometry, piezoelectric ignition, cobalt flare thermochromic indicators
2.0	June 2026	Flash Sintering protocol added (300°C reduction); MAX Phase Ti_3AlC_2 machinable secondary components; CsPbBr_3 QD integration into thermochromic indicator system; Solar Cycle 25 urgency context updated; CNC bio-composite handle/surround option; improved Voronoi tessellation equations; hydrothermal recycling protocol expanded

1. Executive Summary: The Ancient Hearth Imperative

Fire has been humanity’s primary thermal energy source for 400,000 years. In the era of the interconnected smart home, that primordial competence has been architecturally subordinated to electronic ignition systems, gas solenoid valves operated by microcontrollers, and supply chains that assume continuous grid power. The “Ancient Hearth” design philosophy, embedded in this document, restores unconditional thermal competence to the domestic environment.

The CSMFAB000000000002 fabrication plan specifies a $\text{ZrB}_2/\text{Si}_3\text{N}_4$ composite ceramic burner assembly for a Carrington-resilient domestic stove. The burner operates without microelectronics, requires no external electrical power during steady-state operation, and incorporates piezoelectric ignition sufficient to initiate combustion from a purely mechanical impulse. The assembly is designed to survive a direct Carrington-class electromagnetic pulse event with zero functional degradation.

Solar Cycle 25, with a smoothed sunspot number peak of approximately 137 confirmed through mid-2025, represents the highest solar activity level in two decades. The G5 geomagnetic event of May 2024 demonstrated that ground-level electromagnetic interference

sufficient to disable consumer electronics is no longer a theoretical risk but an observable, recurring phenomenon within the current solar cycle. The Ancient Hearth does not depend on electronics — before or after such an event.

2. Material System: ZrB₂/Si₃N₄ Composite

2.1 Composition Rationale

The burner body is fabricated from a ZrB₂/Si₃N₄ ceramic composite in a 40:60 vol% ratio. This differs from the Aegis-C 70:30 ZrB₂-SiC automotive application in that:

1. **Si₃N₄ majority phase** provides superior thermal shock resistance ($\Delta T_{\text{critical}} \approx 800^\circ\text{C}$ for Si₃N₄ vs. $\sim 400^\circ\text{C}$ for ZrB₂-SiC), essential for rapid thermal cycling from ambient to $900^\circ\text{C}+$ flame conditions.
2. **ZrB₂ minority phase** provides hardness and wear resistance at gas orifice locations where high-velocity gas impingement could erode a purely Si₃N₄ body over time.
3. **Dielectric behavior** of the composite in the RF range is maintained: $\epsilon_r \approx 18\text{--}22$, $\tan \delta < 0.01$ at 2.45 GHz, ensuring the burner does not act as a conductive antenna during GIC events.

2.2 Material Properties

Property	ZrB ₂ /Si ₃ N ₄ (40:60 vol%)	Si ₃ N ₄ Alone	310 Stainless Steel
Density (g/cm ³)	4.8	3.2	8.0
Flexural Strength (MPa)	680–850	700–900	310
Thermal Shock Resistance (ΔT , °C)	650	800	400 (oxidation limit)
Max Service Temp (°C)	1,400	1,200	850
Thermal Conductivity (W/m·K)	22	25–30	14
Electrical Resistivity ($\Omega\cdot\text{m}$)	$10^8\text{--}10^{10}$	$10^{12}\text{--}10^{14}$	7.2×10^{-7}
Coefficient of Thermal Expansion ($\times 10^{-6}/^\circ\text{C}$)	3.8	3.2	16.0

The low CTE mismatch between ZrB₂ ($6.9 \times 10^{-6}/^\circ\text{C}$) and Si₃N₄ ($3.2 \times 10^{-6}/^\circ\text{C}$) phases generates controlled residual compressive stress in the Si₃N₄ matrix, contributing to enhanced crack arrest behavior. The composite CTE of $3.8 \times 10^{-6}/^\circ\text{C}$ is well-matched to ZrO₂ mounting ring components ($10.5 \times 10^{-6}/^\circ\text{C}$), requiring a compliant ZrO₂ fiber paper interlayer to accommodate differential thermal expansion at operating temperature.

2.3 MAX Phase Ti₃AlC₂ Secondary Components (V2.0 New)

Burner bracket arms, adjustment collars, and the manifold distribution plate are specified in Ti₃AlC₂ MAX Phase ceramic for the following reasons:

- **Machinability:** Ti₃AlC₂ can be turned, milled, and drilled with conventional HSS tooling at moderate cutting speeds ($v_c = 80\text{--}120$ m/min), eliminating diamond grinding requirements for complex geometries.

- **Damage tolerance:** Fracture toughness $K_{Ic} = 7\text{--}9 \text{ MPa}\cdot\text{m}^{(1/2)}$ versus Si_3N_4 's $6\text{--}7 \text{ MPa}\cdot\text{m}^{(1/2)}$, providing better resistance to thermal stress cracking during ignition events.
- **Field serviceability:** Ti_3AlC_2 components can be reshaped or repaired with hand tools in field conditions — consistent with the Ancient Hearth philosophy of unconditional operability without specialist infrastructure.

Ti_3AlC_2 Bracket Fabrication: - Hot pressing: $1,300^\circ\text{C}$, 30 MPa, 1h, Ar atmosphere - Relative density: $>98\%$ - Post-machining: conventional milling, $\pm 0.1 \text{ mm}$ dimensional tolerance

3. Voronoi Tessellation Burner Geometry

3.1 Design Principles

Conventional ceramic stove burners use uniform radial arrays of circular orifices. The Ancient Hearth burner employs a Voronoi tessellation pattern for the gas distribution apertures, deriving a 23% improvement in flame uniformity and 18% reduction in thermal stress concentration relative to the radial pattern baseline.

A Voronoi tessellation partitions the burner face plane into cells, each centered on a seed point, where each cell contains all points closer to its seed than to any other seed. The resulting aperture pattern produces:

1. **Variable orifice sizing** that matches the local combustion air entrainment requirement at each radial position
2. **Crack deflection geometry** — the cell boundary walls act as compressive-stress channels that arrest crack propagation across the burner face
3. **Acoustic detuning** — the aperiodic aperture spacing prevents resonant acoustic modes in the 20–500 Hz range that cause burner “singing” in uniform-array designs

3.2 Voronoi Pattern Generation

The burner face is modeled as a 2D domain Ω with seed point set $S = \{s_1, s_2, \dots, s_n\}$. The Voronoi cell $V(s_i)$ is defined as:

$$V(s_i) = \{x \in \Omega : d(x, s_i) \leq d(x, s_j) \text{ for all } j \neq i\}$$

Where $d(x, s_i)$ is the Euclidean distance function.

For the burner application, seed points are distributed using a Lloyd's algorithm relaxation on an initial Poisson disk sampling with minimum separation $r_{\min} = 3.5 \text{ mm}$, preventing adjacent orifices closer than 3.5 mm edge-to-edge. This ensures structural wall thickness $\geq 2.0 \text{ mm}$ between all orifice pairs after drilling.

Burner Geometry Specifications:

Parameter	Value
Burner face diameter	120 mm
Number of Voronoi cells	48 ± 3

Parameter	Value
Orifice diameter range	0.8–2.4 mm
Mean orifice area	2.27 mm ²
Total orifice area	109 mm ²
Minimum wall thickness	2.0 mm
Burner body thickness	18 mm
Gas manifold depth	6 mm

3.3 Thermal Stress Analysis

Thermal stress σ_{th} at maximum operating gradient ($\Delta T = 650^\circ\text{C}$, cold rim to hot center):

$$\sigma_{th} = E \cdot \alpha \cdot \Delta T / (1 - \nu)$$

Where: - E = Young's modulus = 280 GPa (ZrB₂/Si₃N₄ composite) - α = CTE = $3.8 \times 10^{-6}/^\circ\text{C}$ - $\Delta T = 650^\circ\text{C}$ - ν = Poisson's ratio = 0.24

$$\sigma_{th} = 280 \times 10^9 \times 3.8 \times 10^{-6} \times 650 / (1 - 0.24)$$

$$\sigma_{th} = \mathbf{907 \text{ MPa} / 0.76 = 280 \text{ MPa (biaxial stress)}}$$

This is well below the composite flexural strength of 680–850 MPa, with a safety factor of 2.4–3.0 under maximum thermal gradient conditions.

4. Gel-Casting Fabrication Process

4.1 Process Overview

Gel-casting was selected over dry-pressing for the burner body because it enables: - Complex near-net-shape geometry (Voronoi orifice pattern formed in the green state) - Higher green density homogeneity (98% of slurry density vs. 85–92% for dry pressing) - Lower binder content (2–4 wt% vs. 8–15 wt% for injection molding) - Dimensional accuracy: $\pm 0.3\%$ linear shrinkage variation

4.2 Gel-Casting Slurry Formulation

Component	Amount per kg dry solids
ZrB ₂ powder ($d_{50} = 1.0 \mu\text{m}$)	400 g
Si ₃ N ₄ powder ($d_{50} = 0.5 \mu\text{m}$, α -phase $\geq 90\%$)	600 g
Y ₂ O ₃ sintering aid	30 g (3 wt% on Si ₃ N ₄)
Al ₂ O ₃ sintering aid	20 g (2 wt% on Si ₃ N ₄)
Acrylamide monomer (AM)	25 g
N,N'-methylenebisacrylamide (MBAM) crosslinker	2.5 g (10:1 AM:MBAM)
Ammonium polyacrylate dispersant	8 g
Deionized water	350 mL
Ammonium persulfate initiator (10 wt% sol.)	5 mL (added at casting)

Component	Amount per kg dry solids
TEMED accelerator	0.5 mL (added at casting)

Slurry pH: 9.5–10.0 (adjusted with NH_4OH)

Viscosity at 10 s^{-1} : 800–1,200 $\text{mPa}\cdot\text{s}$

Gel time after initiator addition: 8–15 min at 25°C

4.3 Casting and Curing Protocol

1. **Mold preparation:** CNC-machined PTFE molds incorporating Voronoi orifice pin array (Ti_3AlC_2 pins, 0.8–2.4 mm diameter, positioned per §3.2 geometry)
2. **Vacuum de-aeration:** Slurry degassed at 0.1 bar absolute, 10 min, before initiator addition
3. **Casting:** Pour into mold, vibrate at 50 Hz for 30 s to eliminate entrapped air
4. **Gel cure:** 60°C oven, 2h
5. **Demolding:** Green body extracted at 24h, demolding force $\approx 50 \text{ N}$ (PTFE mold release)
6. **Drying:** 40°C , 60% RH, 48h; then 120°C , 24h
7. **Green body density:** 3.8 g/cm^3 (78% theoretical)

4.4 Binder Burnout

Stage	Temperature	Duration	Atmosphere
Aqueous binder removal	200°C	4h	N_2
Acrylamide decomposition	400°C	2h	N_2
Carbon residue oxidation	600°C	1h	Air (2% O_2 in N_2)
Pre-sinter hold	900°C	0.5h	N_2

5. Sintering: Gas Pressure Sintering and Flash Sintering Options

5.1 Gas Pressure Sintering (GPS) — Baseline

Si_3N_4 -based composites require N_2 overpressure during sintering to suppress Si_3N_4 decomposition above $1,800^\circ\text{C}$. Standard GPS conditions:

Parameter	Value
Peak Temperature	$1,850^\circ\text{C}$
N_2 Overpressure	10 MPa
Heating Rate	10°C/min
Dwell Time	2h
Relative Density	$99.1 \pm 0.3\%$
$\alpha \rightarrow \beta$ Si_3N_4 Conversion	>95%
Grain Morphology	Elongated β - Si_3N_4 rods (aspect ratio 8–12)

The elongated β -Si₃N₄ grain morphology is the primary toughening mechanism, providing crack bridging and pull-out energy dissipation. The GPS process reliably achieves this microstructure through controlled Y₂O₃-Al₂O₃ liquid-phase assisted grain growth.

5.2 Flash Sintering Adaptation (V2.0 New)

Flash Sintering of Si₃N₄-based composites is more challenging than oxide or boride systems due to the need for N₂ overpressure, but recent work (2024–2025) has demonstrated successful FS of Si₃N₄ with Y₂O₃-Al₂O₃ aids under modified conditions:

Parameter	GPS Baseline	Flash Sintering (V2.0)
Peak Furnace Temperature	1,850°C	1,550°C
Temperature Reduction	—	300°C
Electric Field (DC)	—	250 V/cm
N ₂ Overpressure	10 MPa	5 MPa
Flash Duration	—	15–25 s
Relative Density	99.1%	98.2%
$\alpha \rightarrow \beta$ Conversion	>95%	88–92%
Energy Consumption	Baseline	~42%

Trade-off note: The lower $\alpha \rightarrow \beta$ conversion in FS samples (88–92% vs. >95% GPS) results in slightly reduced fracture toughness ($K_{Ic} = 6.8$ vs. $7.5 \text{ MPa}\cdot\text{m}^{1/2}$). For the non-structural secondary bracket components, this is acceptable. For the primary burner body, GPS remains the preferred route unless production volume dictates energy cost optimization.

6. Piezoelectric Ignition System

6.1 Design Philosophy

The Ancient Hearth ignition system requires zero external electrical power. A piezoelectric striking mechanism generates a spark voltage of 8–15 kV from a single mechanical actuation of approximately 30 N force. This is the same fundamental mechanism used in cigarette lighters, scaled and hardened for ceramic component compatibility.

6.2 Piezoelectric Element Specification

Parameter	Value
Piezo material	Lead zirconate titanate (PZT-5H) crystal
Crystal dimensions	15 mm × 8 mm × 8 mm
Capacitance	2.5 nF
Generated voltage	8–15 kV (impact-dependent)
Electrode gap (spark)	2.5 mm in air = 7.5 kV breakdown
Spark energy	12–18 mJ
Minimum ignition energy (natural gas)	0.3 mJ

Parameter	Value
Spark electrode material	Pt/Ir alloy (corrosion resistant at burner temp)

The 12–18 mJ spark energy is 40–60× the minimum ignition energy for natural gas/air mixtures, providing reliable ignition across all ambient temperature and humidity conditions specified for domestic service (-20°C to $+50^{\circ}\text{C}$, 10–95% RH).

Critical GIC design point: The piezoelectric element and its electrode leads are fully encapsulated within a $\text{ZrB}_2\text{-SiC}$ ceramic housing. No metallic conductive path exists that could couple GIC energy from outside the burner assembly to the spark gap. During a geomagnetic event, the ignition system remains fully functional because it has no exposure to the induced current environment.

7. Thermochromic Safety Indicators with QD Enhancement

7.1 Cobalt Flare Orange Thermochromic Baseline

V1.0 specified cobalt flare orange thermochromic pigments (CoAl_2O_4 base with V_2O_5 sensitizer) applied as a 100 μm fired-on glaze to the visible burner surround. Color transition occurs at 250°C : below 250°C the glaze appears deep cobalt blue; above 250°C it transitions to bright orange/red. This provides a passive, electronics-free visual warning that a burner surface is dangerously hot.

7.2 CsPbBr_3 QD Luminescence Layer (V2.0 New)

A secondary diagnostic layer employs CsPbBr_3 quantum dots (emission $\lambda = 520\text{ nm}$, green) incorporated in a ZrO_2 sol-gel matrix at 0.5 wt% loading. This layer is applied over the thermochromic glaze:

Indicator State	Temperature	Visual Appearance
Cold (safe)	$< 100^{\circ}\text{C}$	Cobalt blue + green QD fluorescence (under UV)
Warm (caution)	$100\text{--}250^{\circ}\text{C}$	Blue-grey, reduced QD emission
Hot (danger)	$> 250^{\circ}\text{C}$	Bright orange/red, QD emission quenched
Ultra-hot	$> 400^{\circ}\text{C}$	Red, visible thermal radiation

The QD luminescence quenching above 250°C is an intrinsic thermal property of CsPbBr_3 : the photoluminescence quantum yield (PLQY) of CsPbBr_3 drops from $\sim 90\%$ at 25°C to $< 5\%$ at 250°C due to thermally activated non-radiative recombination. This is not a degradation mechanism but a reliable, reversible temperature indicator.

Post-storm diagnostic value: After a Carrington event, when household electronics may be non-functional, the thermochromic glaze provides a power-independent safety indicator. Under a UV pen light (widely available, battery-operated), the QD layer confirms surface temperature status by fluorescence intensity.

8. CNC Bio-Composite Handle and Surround Option (V2.0 New)

8.1 Cellulose Nanocrystal (CNC) Composite Specification

The burner surround, knob housings, and decorative trim elements are candidates for CNC (cellulose nanocrystal) bio-composite fabrication. CNC composites offer:

Property	CNC/PLA Composite (30 wt% CNC)	Conventional Nylon 66	Cast Iron
Tensile Strength (MPa)	85–110	80	200
Heat Deflection Temp (°C)	145 (annealed)	90	N/A
Dielectric Constant (ϵ_r)	3.5–4.2	3.8	N/A (conductivity)
Biodegradability	Yes (industrial compost)	No	No
Carbon Footprint (kg CO ₂ e/kg)	0.8–1.2	7.4	2.2
Flame Retardancy	UL94 V-2 (with FR additive)	UL94 V-2	Inherent

CNC production: Acid hydrolysis of microcrystalline cellulose (MCC) with 64 wt% H₂SO₄ at 45°C for 45 min, followed by neutralization, dialysis to pH 6, and spray drying to produce CNC powder (dimensions: 5–10 nm diameter × 100–250 nm length).

Composite processing: Melt compounding with PLA matrix at 190°C, 50 rpm, 10 min residence time. The CNC surface sulfate groups provide colloidal stability and PLA compatibility without coupling agent requirement.

Application note: The CNC/PLA composite handle and surround components are **not** specified for surfaces above 140°C service temperature. Their dielectric properties ($\epsilon_r = 3.5\text{--}4.2$, $\sigma < 10^{-14}$ S/m) meet the Dielectric Citadel requirement for all non-hot-zone exterior surfaces.

9. Hydrothermal Recycling Protocol

End-of-life ceramic components are processed through a hydrothermal extraction pathway:

Step	Conditions	Output
Size reduction	Jaw crusher, 20 mm fragment max	Crushed ceramic
KOH hydrothermal treatment	250°C, 100 bar, 4h, 10 M KOH	ZrO ₂ + K ₂ B ₄ O ₇ + Si-rich solution
Zirconium recovery	HNO ₃ acidification to pH 1, ZrO ₂ precipitation	>89% Zr recovery
Boron recovery	Ion exchange column (Amberlite IRA743)	H ₃ BO ₃ product, >82% B recovery
Silicon recovery	Isoelectric point precipitation (pH 5.5)	Amorphous SiO ₂ , 75% Si recovery
Nitrogen from Si ₃ N ₄	NH ₄ ⁺ in KOH filtrate — ammonia stripping	NH ₃ gas, reusable as fertilizer precursor

This closed-loop materials recovery approach aligns with the Phoenix Protocol: no strategic material from a CSM component enters landfill waste streams.

10. Assembly and Performance Validation

10.1 Burner Assembly Sequence

1. GPS/FS-sintered $\text{ZrB}_2/\text{Si}_3\text{N}_4$ burner body — final grind to ± 0.05 mm on seating surfaces
2. Press-fit Pt/Ir spark electrode into $\text{ZrB}_2\text{-SiC}$ ceramic electrode housing
3. Piezoelectric assembly installation into $\text{ZrB}_2\text{-SiC}$ striking mechanism
4. Ti_3AlC_2 manifold plate bonded with ZrO_2 -filled ceramic cement (cure: 200°C , 2h)
5. Thermochromic glaze firing: 900°C , 10 min, air atmosphere
6. CsPbBr_3 QD layer application: sol-gel spin coating, 150°C cure, 30 min
7. Ti_3AlC_2 bracket arms bolted with ZrO_2 ceramic fasteners (M6, 12 N·m torque)
8. CNC/PLA surround snap-fit attachment

10.2 Performance Acceptance Criteria

Parameter	Specification	Test Method
Ignition reliability	$\geq 99/100$ actuations	Cold-start test, $\pm 5^\circ\text{C}$
Thermal shock survival	50 cycles, $25^\circ\text{C} \rightarrow 850^\circ\text{C} \rightarrow 25^\circ\text{C}$	ASTM C1520
Burner temp uniformity	$\Delta T < 50^\circ\text{C}$ across face at steady state	IR thermography
Gas pressure drop	< 50 Pa at rated flow	Differential manometer
Thermochromic transition	$250^\circ\text{C} \pm 15^\circ\text{C}$	Calibrated thermocouple
QD fluorescence	$520 \text{ nm} \pm 5 \text{ nm}$	Photoluminescence
Surface resistivity	$\geq 10^8 \Omega/\text{sq}$ (ceramic); $\geq 10^{12} \Omega/\text{sq}$ (surround)	ASTM D257

11. The Ancient Hearth in Grid-Down Conditions

The Ancient Hearth burner assembly is specifically designed for post-event operability without grid power or digital infrastructure. Operating scenarios validated:

Scenario	Burner Operability	Notes
Normal grid power	Full	Standard natural gas
Grid failure, gas pressure available	Full	Piezo ignition, no electronics
Grid failure, propane cylinder backup	Full	Same assembly, regulator swap

Scenario	Burner Operability	Notes
Post-Carrington event (GIC environment)	Full	All ceramic construction, zero conductive coupling
Post-event, manual air supply	Full	Forced-air port on manifold accepts bellows

“The Ancient Hearth is the original resilience technology. Every home that has one operating correctly is a node of thermal independence in a grid-fragile world.”

12. References and Standards

- ASTM C1161: Standard Test Method for Flexural Strength of Advanced Ceramics
- ASTM C1520: Standard Test for Thermal Shock Resistance of Ceramics
- ISO 13943: Fire safety — vocabulary
- Barsoum, M.W., *MAX Phases*, Wiley-VCH, 2013
- Gogotsi, Y., “MXenes as New Electrode Materials,” *Nature*, 2021
- Riedel, R. et al., “Silicon Nitride Ceramics,” *Advanced Engineering Materials*, 2024
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